Reference-Free Prediction of Consequential Brain-Tumour Volume Error

Introduction

Gliomas are among the most common and aggressive primary brain tumours, and magnetic resonance imaging (MRI) is the principal modality for their diagnosis, treatment planning, and longitudinal monitoring [1]. A central quantitative output of this imaging is **tumour volume**: changes in the volume of the whole tumour, tumour core, and enhancing region over time are used to assess disease progression and treatment response, and underpin established response-assessment criteria such as RANO, which defines progression in part through a ≥25 % increase in tumour burden [2]. Accurate quantification of these volumes is therefore not merely a segmentation exercise — it directly shapes clinical interpretation.

Deep learning has driven substantial progress in automated brain-tumour segmentation, with benchmarks such as BraTS establishing 3D convolutional architectures (e.g., nnU-Net, SegResNet) as strong, reproducible baselines [3], [4]. In principle these models could automate laborious manual contouring. In practice their outputs cannot be trusted blindly: even high-performing models produce occasional failures, and a single segmentation that substantially mis-estimates tumour volume can propagate into an incorrect assessment of progression or response. Deploying automated segmentation at scale therefore requires more than an accurate model — it requires knowing, for each individual scan, **whether the resulting output can be trusted**, so that unreliable cases are routed to expert review while the rest are safely automated.

Prior work has approached this reliability problem from several directions, and each leaves a specific gap. Uncertainty quantification — Monte-Carlo dropout, deep ensembles, and test-time augmentation — yields voxel-wise uncertainty maps, and the QU-BraTS benchmark showed that such estimates carry information about where a model is likely to err, while also concluding that no single uncertainty method dominates [5]. However, these estimates are evaluated and rewarded mainly at the voxel level. Jungo, Balsiger and Reyes found that uncertainty that is well-calibrated at the dataset level becomes **miscalibrated at the subject level**, with limited ability to localise error, even though aggregating voxel uncertainty to the case level can still flag clearly failed segmentations [6]. This motivates treating case-level reliability as a prediction task in its own right rather than reading it directly off raw uncertainty. Reference-free quality-control methods address the case level — reverse classification accuracy predicts segmentation quality without ground truth by retraining a classifier on the prediction [7], and learned quality predictors flag failures at population scale [8] — but both target **overlap** quality (Dice-like) rather than volume error, are not framed against a pre-specified clinical threshold, and are not reported as an operational triage outcome. More recent work provides calibrated volumetric uncertainty and conformal guarantees: evidential ensembles yield calibrated volume estimates for meningioma segmentation with external validation [9], and conformal methods give predictive intervals for volume that remain valid under covariate shift [10]. These produce an *uncertainty estimate or interval*, not a *decision* about whether a given case is wrong enough to matter clinically. A related line characterises volumetric **bias** of soft-Dice-trained models, but at the population and theoretical level rather than as a per-case prediction [11]. Finally, reliability studies under distribution shift [12] and selective-prediction triage pipelines [13] are emerging, though the former targets Dice-based error detection under synthetic

corruption and the latter addresses tumour *classification* rather than segmentation; and reference-free indicators have in some settings proved insufficient for subtle, non-empty errors [14], a cautionary result that any case-level reliability claim must confront honestly.

Across this literature, three gaps recur: reliability is related to overlap metrics such as the Dice coefficient rather than to **volume error** specifically; error is seldom defined against a **pre-specified, clinically motivated threshold**; and results are rarely reported as an operational **triage** outcome — how many cases can be safely automated at a fixed error-detection sensitivity.

In this work we study segmentation reliability as the **case-level prediction of consequential volume error**. Rather than proposing a new segmentation architecture or a new uncertainty-calibration method, we take a standard segmenter and ask a focused, clinically framed question: can simple, interpretable models predict — from the segmenter's own stochastic behaviour (Monte-Carlo dropout with test-time augmentation) — when an output's whole-tumour volume error exceeds a threshold fixed in advance on clinical grounds? The error endpoint is pre-specified and frozen before any evaluation on the held-out test set, with the commit timestamp as evidence that it was not tuned to the result. From the stochastic predictions we derive case-level features spanning uncertainty, inter-prediction disagreement, and boundary/shape characteristics, and compare feature families against trivial (predicted-volume-only) and single-best-uncertainty baselines using a paired bootstrap. We additionally ask whether the same reference-free signals can indicate the **direction** of the error — whether a case is likely over- or under-segmented — a per-case question distinct from the population-level volumetric bias characterised previously [11]. The predictor is validated against true errors measured from expert reference masks, evaluated on held-out patients as a triage mechanism, and reported with failure cases and negative findings.

We do not claim a novel uncertainty estimator, a calibration method, or a statistical coverage guarantee; those have been studied elsewhere [9], [10]. Nor do we claim that case-level reliability is newly possible — only that it has not been posed in the specific form we adopt. Our aim is a clear, reproducible, clinically-framed baseline for *consequential volume-error triage*, runnable on a single commodity GPU, and an honest account of which readily available signals actually predict it.

Our contributions are:

1. We **formulate segmentation reliability as the case-level prediction of consequential volume error**, with the error endpoint pre-specified against a clinically motivated threshold and frozen before test evaluation, rather than tuned post hoc.
2. We **compare feature families** — uncertainty, prediction disagreement, and boundary/morphology descriptors — against trivial predicted-volume-only and single-best-uncertainty baselines with a paired bootstrap, isolating what actually predicts volume error, and we additionally test whether these signals indicate the **direction** of the error per case.
3. We **evaluate the system with triage-oriented metrics** — AUROC, AUPRC, sensitivity, false-alert rate, the fraction of cases retainable for automation at a fixed sensitivity, and the risk–coverage curve — on held-out patients, reporting failure cases and negative findings alongside successes.

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Methodology

Describes the implemented pipeline (what we did and how), the feasibility experiments run so far, and the planned extensions (what we will do), including the learned end-to-end error detector and the move to a larger training corpus. Completed work is written in the past tense; planned work in the future tense.

1. Problem formulation

We treat brain-tumour segmentation reliability as a **case-level prediction task**: given an automatic segmentation and the model's own stochastic behaviour — but **no reference mask** — decide whether the output's whole-tumour volume error is large enough to be clinically consequential, so that unreliable cases are routed to expert review while the rest are safely automated. The study is deliberately a *triage benchmark*: we do not propose a new segmentation architecture or a new uncertainty estimator, but ask which readily available, reference-free signals actually predict consequential volume error, under a protocol fixed in advance.

2. Data and patient-level splits

All experiments use publicly available, expert-annotated brain-tumour MRI. For the feasibility stage reported here we used the **MSD Task01_BrainTumour** collection (484 patients, four co-registered modalities each). The identical pipeline targets larger glioma corpora (§9) through a configuration switch that carries each dataset's label convention and file layout.

Each patient was assigned to exactly one of four disjoint splits (`scripts/00_make_splits.py`), so no patient appears in more than one:

Split	Share	Purpose
`seg_train`	45 %	train the segmenter only
`seg_val`	5 %	select the segmenter checkpoint; report its Dice
`risk_dev`	30 %	develop risk models (repeated cross-validation), fix thresholds
`risk_test`	20 %	evaluate once, after the protocol is committed

This separation is essential to validity: the segmenter never sees the risk cases, so the errors the risk model learns from are realistic rather than the unrealistically small errors a model makes on its own training data. Datasets that share source scans (e.g. MSD and BraTS) are never combined.

3. Segmentation model

The segmenter is a standard residual 3D U-Net (MONAI) with dropout in every unit, trained once with fixed hyper-parameters (96^3 patches, batch size 1, mixed-precision, Dice loss, Adam, learning rate 2×10^{-4}) and no architecture search. We report its Dice against published BraTS numbers so that reviewers can confirm it is a reasonable, not deliberately weakened, baseline. Dropout in every unit is what makes Monte-Carlo dropout a usable uncertainty source on a single 6 GB GPU, avoiding the cost of a deep ensemble.

4. Reference-free stochastic inference

For each case we drew multiple stochastic predictions with MC-dropout kept active, each pass additionally combined with a random axis flip (test-time augmentation), so that the

passes cost the same as plain MC-dropout. From the per-voxel **mean** and **variance** across passes we computed all downstream signals, for the three nested BraTS regions — whole tumour (WT), tumour core (TC), enhancing tumour (ET). No reference mask enters this stage.

5. Case-level, reference-free features

From the stochastic probability maps we derived per-region features requiring no ground truth, organised into the families that the comparison contrasts:

- `uncertainty` — mean entropy and variance inside the predicted mask and a dilated ROI, and total entropy volume (standard voxel-uncertainty summaries aggregated to the case level);
- `disagreement` — coefficient of variation and relative range of the per-pass predicted volume, and the mean/minimum Dice between each pass and the consensus mask (how much the volume estimate moves across passes);
- `boundary_morphology` — boundary-band (shell) entropy and variance; the *undecided volume fraction* (voxels with probability between 0.2 and 0.8, relative to predicted volume) and the soft-minus-hard volume difference, both direct proxies for how far the volume estimate could move; and shape descriptors (connected components, largest-component fraction, number of small components, surface-to-volume ratio) with log predicted volume;
- `combined` — all of the above across all regions plus nested-region consistency terms (ET outside TC, TC outside WT).

Two reference points anchor every comparison: `volume_only` (log predicted volume alone — small tumours have large relative errors) and `best_single` (the single strongest uncertainty score, selected on `risk_dev`), i.e. the "just threshold one uncertainty number" baseline.

6. Pre-specified error endpoints

The analysis protocol was fixed and committed before any evaluation on the test set (`configs/preregistration.yaml`; the commit timestamp is the evidence, and the evaluation script refuses to run with uncommitted changes to it). The **primary endpoint** labels a case consequential when the whole-tumour relative volume error is ≥ 25 % **and** the absolute error is ≥ 1 mL: the 25 % links to the RANO progression threshold, and the 1 mL floor prevents a sub-millilitre miss on a tiny lesion from being counted. Reference volume error is measured from the expert mask. Secondary endpoints (WT at 10 % and 40 %, TC at 25 %, and WT Dice < 0.80) are reported in full whatever they show and are not used to pick the primary result.

7. Risk models and comparison

The primary model is L2-regularised logistic regression (class-balanced, median imputation, standardised features); shallow histogram gradient boosting is the secondary model. On `risk_dev` we obtained risk scores by repeated (5×) stratified 5-fold cross-validation (out-of-fold scores), which we used to compare feature families, choose `best_single`, and fix the operating threshold at 90 % sensitivity. The frozen models and threshold were then applied **once** to `risk_test`. The pre-specified success criterion was that a candidate family (`boundary_morphology` or `combined`) beat **both** baselines on a paired bootstrap of AUROC; the outcome was reported whether or not it held.

8. Evaluation

We report AUROC (primary metric) and AUPRC shown against prevalence, each with 95 % bootstrap confidence intervals (2,000 resamples). As a triage mechanism we report, at the fixed-sensitivity operating point, the sensitivity actually achieved, the false-alert rate, the fraction of cases retained for automation, and the residual consequential-error rate among retained cases, together with the risk–coverage curve and its area and the Brier score (read with the caveat that a class-balanced logistic model is not calibrated). Feature families are compared with a paired bootstrap of AUROC, reporting the difference, its 95 % CI, and a two-sided p-value.

9. Planned extensions (future work)

The feasibility stage established a working, pre-registered pipeline and an honest baseline comparison. On that feasibility dataset the engineered features did not outperform the simple baselines on the primary endpoint, which motivates the following planned work; none of it alters the frozen endpoint.

1. **Scale to a large training corpus.** End-to-end learning and reliable effect estimates require many more cases than MSD provides. We will run the identical protocol on a large, expert-annotated glioma corpus (candidate: BraTS 2023 Adult Glioma, 1,251 cases, extendable toward the ~2,200-case BraTS 2024 cohort, and optionally pooled with other whole-tumour-annotated collections), giving roughly an order of magnitude more development cases and a lower, more realistic prevalence of consequential error.

1. **A learned, end-to-end error detector.** In place of hand-crafted features we will train a small, heavily-regularised 3D convolutional network that reads the segmenter's uncertainty volumes directly (per-region mean probability, variance, and entropy, resampled to a fixed low-resolution grid) and outputs a single case-level risk score. The network is deliberately tiny (on the order of 10^4 parameters) with dropout, weight decay, and flip augmentation, because positives are scarce. Crucially, it is evaluated under the *same* pre-registered protocol — the same repeated cross-validation on `risk_dev`, the same fixed-sensitivity threshold, and the same paired-bootstrap comparison against `volume_only` and `best_single` — so that "learned" enters as one more family on equal terms. The learned detector is judged only on a fresh, once-opened test split; the existing feasibility test set is not reused.

1. **Cross-fitting for more positives.** Training several segmenters on disjoint folds so that every case receives an out-of-fold prediction roughly doubles the number of error positives available to the risk model, tightening confidence intervals without lowering the clinical threshold.

1. **Direction and magnitude of error.** Beyond the binary flag, we will test whether the same reference-free signals indicate whether a case is over- or under-segmented and by how much, moving from error *detection* toward reference-free volume *correction*.

1. **External validation and calibration.** We will validate on an out-of-distribution cohort (e.g. BraTS-Africa: lower field strength, different population) and add probability calibration so that outputs can be read as risks.

1 **Annotation-noise context.** We will cite measured inter-rater *volume* variability for BraTS and state where the 25 % threshold sits relative to it, since our feasibility results suggest the engineered signals help mainly once the error exceeds the annotation-noise floor (large errors and overlap failures).